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10 CFR 50.73

GNRO2023-00028

November 30, 2023

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1 Licensee Event Report 2023-001-01,
Valve Closure Test Failure Resulting in Condition Prohibited by Technical
Specification

Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
Renewed License No. NPF-29

Attached is Licensee Event Report (LER) 2023-001-01, Valve Closure Test Failure Resulting in Condition Prohibited by Technical Specification. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B), any operation or condition which was prohibited by the plant's Technical Specifications. This supplement is being provided to include that this condition was also prohibited by Technical Specifications section 3.6.3.3 "Drywell Purge Systems" and add detail to the Reportability section of the LER.

This letter contains no new Regulatory Commitments. Should you have any questions concerning the content of this letter, please contact me at 802-380-5124.

Sincerely,

JAH/saw

A handwritten signature in blue ink, appearing to read "JAH", with a stylized flourish extending from the end.

Attachments: Licensee Event Report 2023-001-01

cc: NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

U.S Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

GNRO2023-00028

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Attachment
Licensee Event Report 2023-001-01

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: gira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Grand Gulf Nuclear Station, Unit 1

☒ **050****2. Docket Number**

416

3. Page

1 OF 3

4. Title

Valve Closure Test Failure Resulting in Condition Prohibited by Technical Specification

5. Event Date

Month	Day	Year
6	6	2023

6. LER Number

Year	Sequential Number	Revision No.
2023	- 001 -	01

7. Report Date

Month	Day	Year
11	29	2023

8. Other Facilities Involved

Facility Name	☐ 050	Docket Number
N/A		N/A
Facility Name	☐ 052	Docket Number
N/A		N/A

9. Operating Mode

1

10. Power Level

98

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)**10 CFR Part 20**☐ 20.2203(a)(2)(vi)**10 CFR Part 50**☐ 50.73(a)(2)(ii)(A)☐ 50.73(a)(2)(viii)(A)☐ 73.1200(a)☐ 20.2201(b)☐ 20.2203(a)(3)(i)☐ 50.36(c)(1)(i)(A)☐ 50.73(a)(2)(ii)(B)☐ 50.73(a)(2)(viii)(B)☐ 73.1200(b)☐ 20.2201(d)☐ 20.2203(a)(3)(ii)☐ 50.36(c)(1)(ii)(A)☐ 50.73(a)(2)(iii)☐ 50.73(a)(2)(ix)(A)☐ 73.1200(c)☐ 20.2203(a)(1)☐ 20.2203(a)(4)☐ 50.36(c)(2)☐ 50.73(a)(2)(iv)(A)☐ 50.73(a)(2)(x)☐ 73.1200(d)☐ 20.2203(a)(2)(i)**10 CFR Part 21**☐ 50.46(a)(3)(ii)☐ 50.73(a)(2)(v)(A)**10 CFR Part 73**☐ 73.1200(e)☐ 20.2203(a)(2)(ii)☐ 21.2(c)☐ 50.69(g)☐ 50.73(a)(2)(v)(B)☐ 73.77(a)(1)☐ 73.1200(f)☐ 20.2203(a)(2)(iii)☐ 50.73(a)(2)(i)(A)☐ 50.73(a)(2)(v)(C)☐ 73.77(a)(2)(i)☐ 73.1200(g)☐ 20.2203(a)(2)(iv)☒ 50.73(a)(2)(i)(B)☐ 50.73(a)(2)(v)(D)☐ 73.77(a)(2)(ii)☐ 73.1200(h)☐ 20.2203(a)(2)(v)☐ 50.73(a)(2)(i)(C)☐ 50.73(a)(2)(vii)☐ **OTHER** (Specify here, in abstract, or NRC 366A).**12. Licensee Contact for this LER****Licensee Contact**

Jeff Hardy

Phone Number (Include area code)

802-380-5124

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	N/A	N/A	N/A	Y	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)**15. Expected Submission Date**

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On June 13, 2023, while trending as found periodic data from motor operated valve diagnostic test on 1E61F003B Drywell Purge Inlet Isolation Valve, it was identified that the butterfly valve disc was traveling past the valve seat, resulting in 1E61F003B and Division 2 of the drywell purge and initial vacuum relief subsystem being declared inoperable. 1E61F003A Division 1 drywell purge and initial vacuum relief subsystem was operable during this time. Per Grand Gulf Technical Specifications, two drywell post-Loss-of-Coolant Accident and two drywell purge vacuum relief subsystems shall be operable in MODES 1, 2 and 3. Division 2 drywell purge and vacuum relief subsystem was determined to have been inoperable since October 26, 2017.

There were no consequences to the general safety of the public, nuclear safety, industrial safety or radiological safety. No radiological releases occurred due to this event.

This report is made pursuant to 10 CFR 50.73 (a)(2)(i)(B), any operation or condition which was prohibited by the plant's Technical Specifications.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Grand Gulf Nuclear Station, Unit 1	☒ 050	2. DOCKET NUMBER 416	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER - 001	REV NO. - 01

NARRATIVE**Plant Conditions:**

Mode 1, 98 percent power

Description of Event:

On October 26, 2017, while performing surveillance for Division 2 post- Loss-of-Coolant Accident (LOCA) Drywell Vacuum Breaker operability, the operator did not observe any movements from the stem of the valve 1E61F003B Drywell Purge Inlet Isolation Valve upon initiation. The control room indications were indicating that the valve was opened and closed. Movements of the stem are required for Technical Specification Acceptance Criteria. Thus, the surveillance was completed unsatisfactory. The valve 1E61F003B and Division 2 Drywell Purge/Initial Vacuum Relief was declared inoperable.

Upon inspection, the spline adapter of valve 1E61F003B was found slipping down the valve stem, causing the disengagement between the valve stem and the drive sleeve. Thus, the actuator was moving and indicating opened/closed. However, the valve didn't move.

On October 27, 2017, the spline adapter was reinstalled per a work order and surveillance was re-performed satisfactorily. This included a current and switches only diagnostic test which was the current test standard for limit controlled soft seated butterfly valves.

On January 24, 2019, it was identified that the diagnostic testing on motor operated valve (MOV) butterfly valves did not currently meet the requirements of GL 96-05/JOG program. This was due to not using a strain gauge to measure valve stem torque. The preventative maintenance requirements (PMRQs) and model work orders (MWOs) were updated for all applicable butterfly valves to reflect that the diagnostic testing required measuring the valve stem torque.

On June 6, 2023, the periodic 1E61F003B MOV diagnostic test was performed with strain gauge on the stem per work order. The test data was within the criteria in the motor operator valve data record.

On June 13, 2023, while trending as found periodic data from the MOV diagnostic test on 1E61F003B, it was identified that the butterfly valve's disc is traveling past the valve seat, resulting in valve 1E61F003B and Division 2 Drywell Purge/Initial Vacuum Relief being declared inoperable. The test data indicates that the valve is contacting the seat at 40 secs of the full 60 sec stroke time. After contacting the seat, the seating torque decreases (relaxes) and the valve continues to travel for 20 secs past the seat. This MOV design has a full travel of 90 degrees. With the actuator in the full open position, the valve disk is 60 degrees open. Conversely with the actuator in the full closed position, the valve disk is 30 degrees open. This valve has no torque switch in the circuit and is seated using the limit switches.

On June 16, 2023, the actuator/valve assembly was adjusted to the proper required position per work order. This was followed by a satisfactory as left diagnostic test using strain gauge to measure valve stem torque. The diagnostic test indicated that the valve disk is now going to full closed position when the actuator is stroked to the full close position.

**LICENSEE EVENT REPORT (LER)
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1. FACILITY NAME Grand Gulf Nuclear Station, Unit 1	☒ 050	2. DOCKET NUMBER 416	3. LER NUMBER		
	☐ 052		YEAR 2023	SEQUENTIAL NUMBER - 001	REV NO. - 01

NARRATIVE**Reportability:**

This was a condition that is prohibited by technical specifications and is reportable per 10 CFR 50.73 (a)(2)(i)(B). Per Grand Gulf Technical Specifications section 3.6.5.6, two drywell post-Loss-of-Coolant Accident (LOCA) and two drywell purge vacuum relief subsystems shall be operable in MODES 1, 2 and 3. This condition was also prohibited by Technical Specifications section 3.6.3.3 "Drywell Purge Systems." Two drywell purge subsystems shall be operable in MODES 1 and 2. Division 2 drywell purge and vacuum relief subsystem was determined to have been inoperable since October 26, 2017.

Cause(s):

Direct Cause: 1E61F003B valve disc traveling past the valve seat was caused by the splined adapter not being properly installed in the past per work order on October 27, 2017. The issue was corrected by adjusting actuator/valve assembly to the proper required position in a later work order.

Corrective Actions:

1. Update procedure to include confirmation of valve stem orientation in relation to the disc orientation.
2. The PMRQs and MWOs were updated for all applicable butterfly valves to reflect that the diagnostic testing required measuring the valve stem torque as part of the corrective actions.

Safety Significance:

The drywell purge system consists of two 100 percent subsystems that serve to purge the hydrogen produced after a LOCA into the larger containment volume for dilution. It also provides a means to relieve drywell vacuum following a LOCA. The 1E61F003B is a Division 2 drywell purge compressor system isolation valve. If the 1E61F003B is open, the drywell purge compressor will be able to pressurize the drywell with Division 1 unavailable. If the 1E61F003B is open and Division 2 is failed, the check valve 1E61F002B will prevent reverse flow.

The safety significance of this event was minimal. Since operation of only one of the two subsystems are required, a single failure will not prevent the system from fulfilling its design function. Therefore, the consequences of this event had minimal impact on the health and safety of the public and reactor safety.

Previous Similar Events:

A review of the corrective action program for the past three years did not identify any similar events.